

SurgFlow: 3D Object-Centric Contact Flow for Surgical Manipulation

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Abstract—Paired video–action demonstrations enable autonomous surgical behavior, but such data is scarce: robots perform roughly 1% of the world’s surgeries, while video-only data is orders of magnitude more abundant. Learning 3D object flow offers an embodiment-agnostic way to use this video, but flow alone specifies how an object should move, not where and when the tool should engage it, a distinction that is critical in surgery. We introduce SurgFlow, a framework that learns 3D *Object-Centric Contact Flow* from stereo surgical video without robot action labels: for each object point, the representation pairs a future 3D trajectory with a contact score, jointly encoding object motion and the tool contact required to produce it. We extract these targets from video by 3D tracking and tool–object proximity, train a flow matching generator to predict them, and control robots by converting predicted contact into grasp and release commands and predicted flow into end effector motion through optimization. On the *da Vinci Research Kit (dVRK)*, SurgFlow succeeds in 37 of 39 stage evaluations across tissue retraction, bimanual reveal, needle pickup, and needle handover, outperforming baselines trained with or without action labels, and transfers zero-shot to the *LapSurgie* platform with average accuracies of 85% and 70% in similar and novel view test cases.

I. INTRODUCTION

Recent behavioral cloning methods, such as SRT-H [1], have demonstrated that paired video–action demonstrations provide effective supervision for autonomous behavior in robotic surgeries. Yet such data is scarce by construction: roughly 99% of the world’s surgeries involve no robot and therefore produce no action labels, and even within robotic surgery kinematics are rarely recorded or released; the largest public action-aligned corpus, assembled from 119 datasets, totals 780 hours [2]. By contrast, individual video-only corpora already exceed this scale (938 hours in LEMON [3]), and every laparoscopic or open procedure is a potential source of more.

Learning from video alone has therefore become an active direction in both general and surgical robotics. Video generation methods treat synthesized frames as a plan [4] but still require an inverse dynamics model trained on paired robot data, and pixel-level prediction couples motion with texture and lighting, making models costly and prone to hallucination [5]. A more compact alternative is to predict motion directly as 3D flow [6], [7], which is viewpoint-invariant and grounded in scene geometry; object-centric variants [8] track flow only on the manipulated object and are embodiment-agnostic, so human demonstrations can

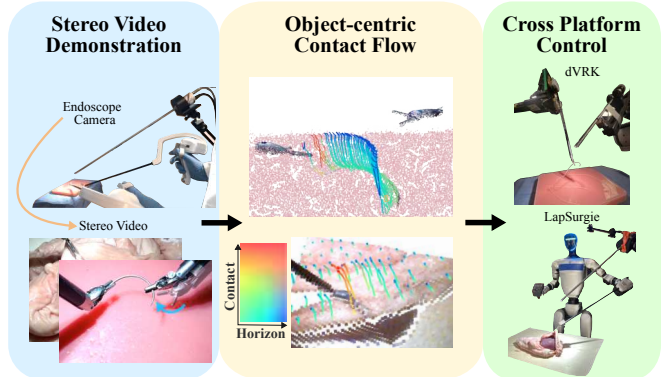


Fig. 1: SurgFlow extracts object motion and tool–object contact from stereo video demonstrations without robot action labels; the resulting contact flow directly guides manipulation on two surgical robot platforms, dVRK and LapSurgie.

supply targets that any robot can execute. However, object flow specifies how the object should move, not how the tool should move it; prior methods leave the interaction point implicit and defer it to heuristics or a separate grasp module. This is tolerable for rigid pick-and-place but not for surgical tasks, where the location and timing of tool–object interaction largely determine success [9], [10]. This motivates our research problem: can we learn from video an object-centric representation that captures both object motion and the tool–object contact required to produce it?

We introduce SurgFlow, a framework that learns 3D *Object-Centric Contact Flow* from stereo surgical video without any robot action labels. For each query point on the object, the representation pairs a future 3D motion trajectory with a contact score: the flow specifies how the object moves or deforms, and the contact specifies where and when the tool should engage, hold, or release it as that motion unfolds. We extract this representation offline from stereo video by reconstructing and tracking object points in 3D and inferring contact from tool–object proximity, and train a flow matching generator to predict future flow and contact from recent observations. Because predictions are expressed in the object frame rather than robot actions, a calibrated controller on any platform can turn contact into grasp and release commands and solve for end-effector motion reproducing the predicted flow, allowing a single learned model to drive different surgical robots. We demonstrate ex-vivo tissue retraction, bimanual reveal, and needle pickup and handover on the dVRK, with zero-shot transfer of the retraction tasks to LapSurgie under viewpoint changes.

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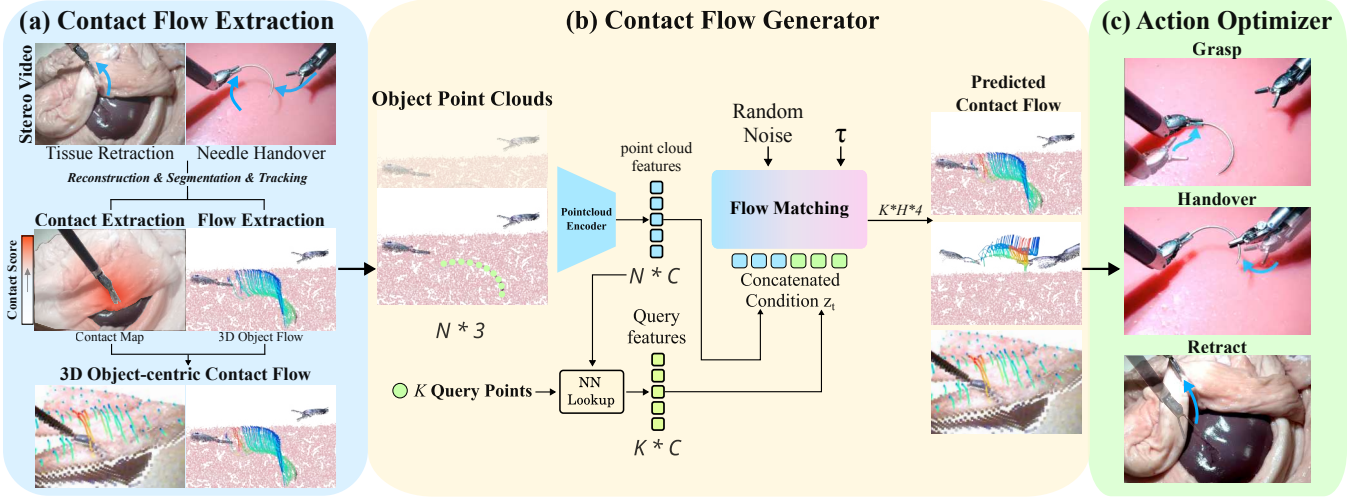


Fig. 2: **Overview of SurgFlow.** (a) Tool–object contact and 3D object flow are extracted from stereo video and combined into an object-centric contact flow representation. (b) Point cloud features are concatenated with query features to condition a flow matching generator that predicts future contact flow. (c) Contact flow supports bimanual manipulation of rigid and deformable objects, from sustained contact in tissue retraction to coordinated grasp and release in needle handover.

II. METHOD

A. Problem Formulation

SurgFlow learns to generate *Object-centric Contact Flow*, which jointly encodes object motion and tool–object interaction, from stereo surgical videos without requiring robot action labels. A canonical object frame with identity orientation and origin at the centroid of the object’s initial point cloud is defined prior to manipulation, and, assuming a well calibrated system, all 3D quantities are expressed within it. For a query point $\mathbf{q} \in \mathbb{R}^3$ sampled on the initial object, its contact flow from t to $t + h$ is expressed as

$$y_{t,h}(\mathbf{q}) = (\mathbf{q}_{t+h} - \mathbf{q}_t, c_{t+h}), \quad (1)$$

where the flow is the smoothed displacement of the query point and $c \in [0, 1]$ is a contact score that indicates whether the object region represented by $\mathbf{q}$ is involved in tool contact.

SurgFlow separates object motion and contact prediction from robot action planning. Its first component, a contact-flow generator f_θ based on flow matching [11] (Sec II-C), is conditioned on the current and historical point cloud observations P_{t-1}^o, P_t^o to predict contact flow $\mathcal{Y}_t = \{y_{t,h}(\mathbf{q}^i)\}$ for a set of query points $Q_t = \{\mathbf{q}^i\}_{i=1}^K$ over a horizon of H steps. The process of extracting contact flow from raw surgical videos is discussed in Sec II-B. The second component converts generated contact flow into the next robot action (Sec II-D); it is implemented as an embodiment-specific controller that yields end effector motions and jaw commands based on optimization and gating. The deliberate separation of the two components allows the same object-centric representation to be used across multiple surgical robotic platforms. Our pipeline is visualized in Fig. 2.

B. Flow and Contact Extraction

A suite of computer vision algorithms — semantic segmentation [12], stereo disparity estimation [13], and video point tracking [14] — are employed to extract object-centric

contact flow from raw video demonstrations. Initial query points $\hat{Q}_0$ are sampled within the object segmentation mask via farthest point sampling.

Flow Extraction. The initial query points are tracked across frames, lifted into 3D via inverse projection using known camera intrinsics and estimated stereo disparity, and smoothed with cubic B-spline fitting to suppress observation noise; the flow of a query over horizon h is the displacement between its smoothed 3D positions at $t + h$ and t .

Contact Extraction and Filtering. Contact is estimated from jaw closure (keypoints on the two jaw branches falling within a distance threshold) and the proximity of each query to the jaw midpoint, quantified as a spatial evidence score that is maximal within a plateau radius and decays exponentially beyond it. Because raw evidence is noisy, the contact value is temporally filtered: while the jaws are closed it is updated toward the current evidence when the query is visible with valid depth and held during occlusion, and it decays smoothly to zero once the jaws open. The filtered scores are paired with the extracted flow to form the training targets in Eq. (1); for bimanual demonstrations, contact from both tools is represented as a single field over the object without tool-specific labels.

C. Contact Flow Generation

An observation encoder maps the previous and current object point clouds and the query set to conditioning tokens $\mathbf{z}_t$, and the generator defines a conditional distribution $\mathcal{Y}_t \sim f_\theta(\cdot | \mathbf{z}_t)$: following conditional flow matching [11], a velocity field is trained to transport Gaussian noise to the per-channel standardized targets along linear interpolation paths. At inference, the learned ODE is integrated from noise with a fixed-step Euler scheme and the standardization inverted, yielding the prediction $\hat{\mathcal{Y}}_t$ passed to the controller.

The encoder is a shared PointNet++ backbone [15] whose per-point features are interpolated at the query locations, and

the velocity field is a DiT-style transformer [16] with one token per query. During training, invalid or occluded targets are excluded in the loss, and rotation augmentation together with random masking of local point cloud patches simulates viewpoint changes and partial observations.

D. Contact Flow to Robot Action

The predicted contact flow $\hat{\mathcal{Y}}_t$ is converted into gripper commands and end effector motions through a combination of gating and optimization.

Contact to Gripper Action. A grasp is initiated at the first time t when any predicted contact value exceeds a threshold τ_{on} . Qualifying queries are grouped by fixed-radius ball clustering, clusters are assigned to arms by proximity, and within each cluster the point q_t^{max} of highest contact is selected as the grasp point, with orientation set by object geometry: opposite the local surface normal for tissue, and perpendicular to the needle plane for handover, following [9]. The gripper approaches q_t^{max} with jaws open, closes on arrival, and releases when the cluster’s maximum contact value falls below τ_{on} .

Flow to End Effector Motion. When the object is grasped, the controller solves inversely for a chunk of end effector motions that best reproduce the predicted object flow near the grasped region, each arm using only queries in its assigned contact cluster. For each future step $t + h$ within the chunk, we solve

$$\Delta R_{t+h}^*, \Delta \mathbf{u}_{t+h}^* = \arg \min_{R, \mathbf{u}} \sum_{i=1}^K w_{t+h}^i \left\| \mathbf{p}_t + \mathbf{u} + R(\mathbf{q}_t^i - \mathbf{p}_t) - \mathbf{q}_{t+h}^i \right\|_2^2, \quad (2)$$

where $\Delta R_{t+h}^* \in SO(3)$ and $\Delta \mathbf{u}_{t+h}^* \in \mathbb{R}^3$ are the optimal end effector rotation and translation relative to time t , and the weight w_{t+h}^i normalizes the predicted contact scores across queries so that points with higher predicted contact have greater influence. The resulting pose targets are executed in sequence before a new prediction is obtained.

III. EXPERIMENTS

A. Experimental Setup

Experiments use two bimanual research surgical robot platforms with stereo perception, dVRK and LapSurgie; quantitative evaluation is primarily on the dVRK, with LapSurgie demonstrating cross-platform transfer.

Evaluation covers four tasks with various contact requirements, material type, and bimanual coordination: **Single Reveal**, where one tissue forcep lifts intestinal tissue until the underlying liver is exposed; **Bimanual Reveal**, where two needle drivers retract separate regions of a phantom shell in opposing directions until the phantom tumor is fully visible; **Needle Pickup**, where a needle driver grasps a rigid surgical needle and transports it to a target location with a stable hold; and **Needle Handover**, continuing from pickup, where Arm 1 presents the needle, Arm 2 grasps it at a location suitable for subsequent suturing, and Arm 1 releases, with success requiring a stable hold at the required location after release.

TABLE I: Task success on the dVRK over shared randomized initial configurations. Handover is evaluated conditionally on successful pickup; DP3 uses 100 action-labeled demonstrations. Best in bold; “—” denotes an inapplicable setting.

Method	Single Reveal	Bimanual reveal	Needle pickup	Needle handover
DP3	9/10	8/10	8/10	4/8
Object Flow(w/o contact)	9/10	4/10	6/10	—
Tool Flow	9/10	7/10	8/10	0/8
SurgFlow+IDM	10/10	10/10	9/10	4/9
SurgFlow	10/10	10/10	9/10	8/9

B. Task Execution and Contact Localization

We compare SurgFlow with baselines that differ in representation and action generation: **Object Flow**, which removes contact prediction and grasps where predicted flow magnitude is largest, following FlowBot3D [6] (denoted **Max Flow** in localization), while retaining the action optimizer; **Tool Flow**, following ToolFlowNet [7], where grasp locations and motions are determined by predicted tool flow, with grasp orientation estimated from object geometry as in SurgFlow and proximity-triggered closure since it does not address jaw commands; **SurgFlow+IDM**, which replaces the action optimizer with an inverse dynamics model trained using robot action labels; and **DP3** [17], which directly predicts robot actions from 3D observations using action-labeled demonstrations.

As shown in Table I, all methods handle single arm retraction, where one persistent grasp makes object motion a sufficient signal, but performance diverges as the required contact structure grows more complex. Object Flow fails on bimanual reveal because flow magnitude does not identify two distinct contacts, while Tool Flow, DP3, and SurgFlow+IDM complete pickup but degrade during handover, which demands an ordered contact transition—receiver grasp before provider release—that their representations leave unspecified. SurgFlow remains reliable across tasks by coupling desired object motion with explicit spatial and temporal contact information.

To evaluate contact localization independently of the controller, we additionally compare predicted grasp points with human expert annotations on 40 examples excluded from training, spanning ten object configurations for each of the four decision states, with error measured as Euclidean distance for the tissue tasks and distance along the needle arc for the needle tasks. SurgFlow, selecting the query with the highest predicted contact score (clustered into two groups for bimanual reveal), achieves the lowest overall error: a median of 3.3 mm versus 4.6 mm for Tool Flow (nearest object point when the rolled-out tool comes within a threshold), 10.4 mm for Max Flow, 13.7 mm for a VLM baseline (Claude Opus 5 predicting an image point lifted to 3D), and 15.9 mm for random sampling, significant over every baseline under one sided paired Wilcoxon tests with Holm correction ($p_{\text{Holm}} \leq 0.007$); qualitative examples are shown in Fig. 3. Max Flow degrades whenever flow

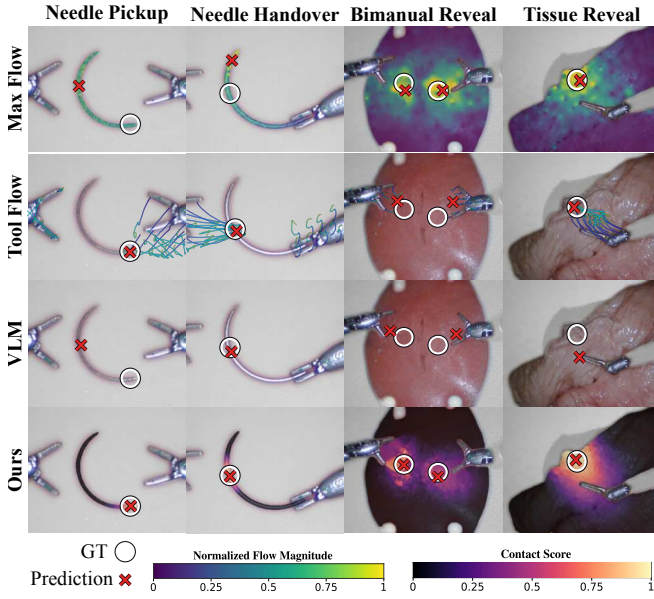


Fig. 3: Predicted grasp locations (red crosses) versus expert annotations (white circles) across the four tasks (columns) for Max Flow, Tool Flow, VLM, and SurgFlow (rows).

TABLE II: Task success on LapSurgie out of ten attempts; the ablation uses SurgFlow trained without occlusion augmentation.

Task	SurgFlow Similar view	SurgFlow Extreme view	w/o aug. Extreme view	DP3
Bimanual reveal	7/10	5/10	1/10	0/10
Ex vivo retraction	10/10	9/10	3/10	0/10

magnitude and desired contact decouple, most severely on needle pickup (18.3 mm vs. 1.3 mm), while Tool Flow infers contact indirectly from the predicted tool trajectory and is sensitive to errors accumulated during rollout.

C. Transfer and Generalization

We evaluate zero-shot transfer to LapSurgie, a calibrated platform with a different base configuration from the dVRK, on single arm ex-vivo tissue retraction and bimanual reveal: the pretrained generator and action optimizer are deployed without fine-tuning under two camera views, one similar to training and one more extreme with greater self-occlusion, and compared against **DP3** transferred from dVRK training and against **SurgFlow w/o augmentation**, trained without occlusion augmentation and tested under the extreme view. As reported in Table II, SurgFlow transfers without fine-tuning (most failures occur during reaching, where LapSurgie’s controller cannot position the tools accurately enough) while the dVRK-trained DP3 policy completes neither task; success drops moderately under the extreme view and sharply without augmentation, indicating that the object-centric frame and augmentation together provide viewpoint robustness.

IV. LIMITATIONS AND CONCLUSION

We demonstrated learning surgical manipulation from stereo video without robot action labels, executed on dVRK

and LapSurgie; jointly representing object motion and contact lets one learned representation guide motion and grasp-release coordination across platforms. Limitations remain: proximity-based contact estimation may not always reflect actual engagement, grasp orientation still relies on geometric heuristics, and calibration and controller errors can prevent reproducing the desired motion, especially on a new platform with less precise control. We see this work as a step toward making greater use of surgical video for robot learning.

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